Review of Diode Forward Voltage-Based Temperature Measurement in Inverter-Like Operation for SiC MOSFETs

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***Abstract*—Accurate virtual junction temperature (T_{vj}) monitoring is essential for lifetime estimation and reliability assessment in Silicon Carbide (SiC) MOSFETs. This review evaluates temperature-sensitive electrical parameters (TSEPs), with emphasis on the third-quadrant diode forward voltage (V_{SD}) method, which offers strong linearity and reduced sensitivity to gate-drive noise. Reliable implementation requires a low sensing current of 50–100 mA and a negative gate bias of -6 to -7 V to suppress bias temperature instability and bipolar degradation. A comparison of measurement topologies reveals a clear trade-off: hardware-assisted schemes achieve sub-Kelvin accuracy with commutation delays of about 100–120 μs, while sensorless dual-gate-bias (DGB) methods avoid added hardware but typically incur 5.0°C–7.5°C error. The review also shows that conventional V_{SD} sensing, with a measurement delay limited to about 10 ms by commutation and current-source settling constraints, misses much of the rapid transient heating and cooling, leading to an underestimation of the peak junction temperature. Using the Foster-network model to quantify this effect, fast 100 μs hardware-based V_{SD} sampling captures the transient response more faithfully.**

***Index Terms*—SiC MOSFET, diode forward voltage, junction temperature measurement, body diode, power cycling, BTI, bipolar degradation, thermal modeling, TSEP.**

I. Introduction

Silicon Carbide (SiC) MOSFETs are increasingly used in high-power converters, where accurate knowledge of the virtual junction temperature (T_{vj}) is essential for reliability and lifetime prediction [7], [8], [13]. Because direct physical or optical temperature measurement is often impractical in packaged devices, temperature-sensitive electrical parameters (TSEPs) are commonly used instead [11], [12]. Among these, the third-quadrant diode forward voltage (V_{SD}) is especially attractive because it offers a strong temperature dependence and can be measured through existing terminals with relatively low hardware complexity [6], [9].

However, practical V_{SD} sensing is constrained by measurement noise, parasitic ringing, and degradation mechanisms such as bias temperature instability and bipolar degradation [10], [16]. The sensing window must therefore be carefully chosen, and the measurement current and gate bias must remain within safe limits to avoid disturbing the device under test [1], [2]. These constraints become more critical when transient heating is fast, because a conventional V_{SD} sensing scheme's multi-millisecond measurement delay can miss the true peak junction temperature and lead to optimistic lifetime estimates [14], [15].

This review compares the main TSEP options, summarizes the operating conditions required for reliable V_{SD} tracking, and evaluates representative online measurement topologies for inverter-like operation. It also examines how measurement delay affects the reconstructed temperature, especially under dynamic active power-cycling(dAPC) conditions where parasitic inductance and capacitance can produce significant transient heating [3]–[5]. The goal is to provide practical guidance for selecting robust V_{SD}-based temperature-monitoring strategies that better reflect actual operational stress in converter systems.

II. Temperature-Sensitive Electrical Parameters and the Diode Forward-Voltage Method

A. Overview of Temperature-Sensitive Electrical Parameters

For packaged SiC MOSFETs, direct physical or optical temperature measurements are often impractical, so electrical temperature-sensitive parameters (TSEPs) are used instead [7], [8]. Commonly employed TSEPs include the on-state resistance $R_{\mathrm{DS(on)}}$, the gate threshold voltage V_{th}, dynamic switching delay times, and the third-quadrant diode forward voltage V_{SD} [2], [11]–[13]. Each of these parameters exhibits a temperature dependence that can be exploited for virtual junction temperature estimation after a suitable calibration procedure.

B. Limitations of Alternative TSEPs

The on-state resistance $R_{\mathrm{DS(on)}}$ monitors the forward voltage drop under high conduction current, but its temperature behavior in SiC MOSFETs is highly non-linear due to competing negative temperature coefficient contributions from the inversion channel and positive temperature coefficient contributions from the drift region, and it is strongly affected by packaging degradation such as bond-wire lift-off [8], [11]–[13]. The gate threshold voltage V_{th} decreases with temperature and

is highly sensitive to bias temperature instability and device-to-device process variations, which complicates its use as a robust online thermometer [2], [12]. Dynamic TSEPs based on switching transients, such as turn-off delay time or dv/dt and di/dt, require high-bandwidth measurement hardware and are strongly influenced by gate-driver and layout parasitics, making them difficult to implement in practical converter systems [3], [14].

C. Diode Forward-Voltage Method and Its Advantages

The third-quadrant source-to-drain forward voltage drop (V_{SD}) across the intrinsic p–n body diode has emerged as a particularly attractive TSEP for Silicon Carbide (SiC) MOSFET diagnostics. Unlike first-quadrant parameters, it is measured when the MOSFET channel is off, so the measurement path is largely decoupled from gate-drive noise and channel conduction effects.

Driven by a constant low sensing current I_{m}, the V_{SD} interaction is described by the ideal diode equation [9]:

$$I_{\mathrm{m}} = I_{\mathrm{s}}(T_{\mathrm{vj}}) \left[\exp\left(\frac{\mathrm{q} V_{\mathrm{SD}}}{\mathrm{k} T_{\mathrm{vj}}} \right) - 1 \right] \tag{1}$$

where I_{s} is the temperature-dependent saturation current, q is the elementary charge, and k is Boltzmann's constant. Differentiating this expression yields an approximately linear temperature coefficient α_V. Experimental studies report sensitivities on the order of $\alpha_V \approx -1.5$ to -2.2 mV/°C over a typical junction temperature range of 25°C–175°C for SiC MOSFETs [6], [9], [12]. This predictable behavior allows the virtual junction temperature to be extracted via a simple first-order linear calibration:

$$T_{\mathrm{vj}} = \frac{1}{\alpha_V} \left(V_{\mathrm{SD}}(T_{\mathrm{vj}}) - V_{\mathrm{SD},0} \right) + T_0 \tag{2}$$

where $V_{\mathrm{SD},0}$ is the calibrated forward voltage drop at the reference temperature T_0, and α_V is treated as a constant within the specified operating range [6].

The main limitations of the V_{SD} method are that it provides only a volume-averaged temperature across the active p–n junction area, without spatial resolution of local hot-spots [7], [12], and that it is susceptible to inductive ringing if the sensing instant is not properly gated [1].

III. Operational Principles of the V_{SD} Method

In inverter-like operation, fast third-quadrant V_{SD} sensing is performed during brief diagnostic windows that follow current commutation into an auxiliary path and precede significant junction cooling [1], [2].

Practical implementations therefore constrain the sensing current to about 50–100 mA (roughly 0.1% of the power-cycling current) and a current density near 60 mA/cm^2 to avoid self-heating and bipolar degradation, while applying a negative gate bias of approximately -6 to -7 V to ensure that the body diode carries the entire sensing current [5], [6], [8], [10], [11]. Hardware-assisted freewheeling topologies reduce commutation tails to roughly 100–120 μs, enabling V_{SD} sampling close to the true peak junction temperature, whereas firmware-based schemes operate directly within sub-microsecond dead times without adding loop inductance but are more exposed to residual RLC ringing [1], [11].

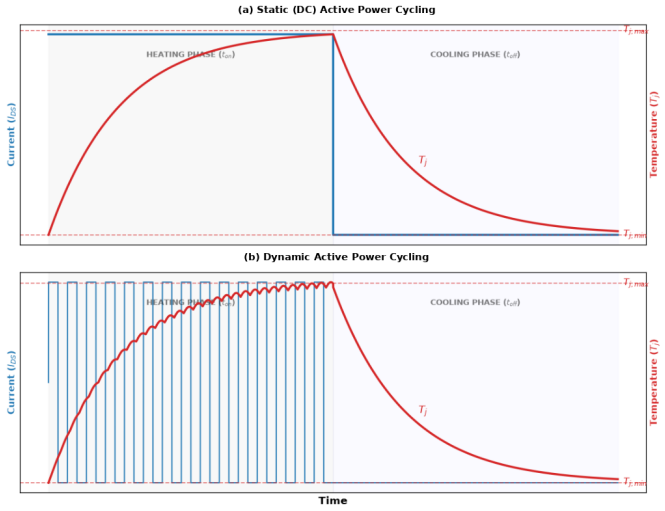


Fig. 1. Comparison of active power cycling profiles: Static (DC) Active Power Cycling (a) versus Dynamic Active Power Cycling (b), depicting current I_{DS} and junction temperature T_{j} over time.

IV. Application of the Method in Power Cycling (AC vs. DC)

Static DC power cycling generates primarily conduction losses and offers wide millisecond-scale sampling windows, but by the time the body-diode voltage is measured the junction has cooled substantially [7], [8]. Dynamic PWM-based cycling adds switching losses and restricts V_{SD} sensing to microsecond-scale dead-time intervals, so the diagnostic window must be tightly aligned with the end of the conduction phase [1], [4], [5]. Fig. 1 contrasts the resulting current and temperature waveforms for static (DC) and dAPC testing. Dynamic active power cycling is therefore more representative of inverter-like operation because it reproduces the combined conduction and switching stress seen in practical converters.

V. Error Sources and Degradation Effects

The main error mechanisms for third-quadrant V_{SD}-based temperature estimation are bias temperature instability, parasitic RLC elements, and stacking-fault-induced bipolar degradation of the body diode [2], [8], [10].

Bias temperature instability (BTI) driven threshold-voltage shifts on the order of 0.1–0.5 V necessitate a sufficiently deep negative gate bias and periodic calibration updates to prevent hybrid channel-plus-diode conduction during sensing [2], [6], [11]. Because the DGB scheme alternates between two negative gate-bias levels during dead time rather than holding a single deep bias throughout, as hardware based sensing does, it is expected to be comparatively more exposed to BTI-driven drift, reinforcing the need for the periodic recalibration noted for firmware-based schemes [11]. Parasitic inductances and

capacitances enforce a blind interval immediately after turn-off; without auxiliary commutation paths, this delay can reach several milliseconds and cause temperature underestimations exceeding 10°C [1], [4]. Heavy third-quadrant conduction at high current density can permanently increase V_{SD} by tens of millivolts, corresponding to more than 10°C error if a fixed linear calibration is used, so sensing schemes restrict current density and pulse width to keep bipolar degradation negligible over the device lifetime [8], [10].

TABLE I
SUMMARY OF V_{SD} MEASUREMENT IMPLEMENTATIONS

Metric	OVMC + Q_{fw}	DGB Scheme
Layout impact	Extra switches, branch	No extra hardware
Added L_{σ}	$\approx$ 2–5 nH	0 nH
Commutation delay	$\approx$ 100–120 μs	$<$ 1 μs window
Temp. error	$< \pm 2 - 3^{\circ}$C	$\approx$ 5–7.5°C
BTI susceptibility	Low (fixed bias)	Higher (bias switching)
Main limitation	Extra BOM, volume	Higher noise sensitivity

OVMC: On-state voltage measurement circuit; Q_{fw}: auxiliary freewheeling switch; DGB: dual-gate-bias; BOM: bill-of-materials.

VI. Testing Topologies and Implementations

Table I summarizes representative implementations of third-quadrant V_{SD} sensing in inverter-like operation, contrasting a hardware-assisted configuration based on an on-state voltage measurement circuit (OVMC) plus an auxiliary freewheeling path Q_{fw} with a purely firmware-based dual-gate-bias (DGB) scheme [1], [2], [11].

A. Hardware-Assisted OVMC + Freewheeling Path

In the OVMC + Q_{fw} approach, dedicated sensing hardware and an auxiliary commutation branch are added to the converter to create a low-noise Kelvin measurement path and to rapidly divert the load current away from the device under test during diagnostic windows [1], [2]. As indicated in Table I, this topology introduces a small additional stray inductance and extra components, but it significantly shortens commutation delays and achieves sub-Kelvin temperature accuracy at the expense of increased bill-of-materials cost and PCB area. Detailed circuit-level descriptions and stability analyses are provided in [1], [2] and are not repeated here.

B. Firmware-Based Dual-Gate-Bias Scheme

The DGB scheme embeds the V_{SD} diagnostic entirely in the intelligent gate driver, using two negative gate-bias levels during dead time to switch between mixed channel-plus-diode conduction and purely diode conduction without any additional hardware [11]. As summarized in Table I, this method adds no extra loop inductance and operates within sub-microsecond windows, but it exhibits a larger temperature error band due to residual layout and gate-driver noise. Its main advantage is the zero incremental hardware overhead, which makes it attractive for volume-constrained traction inverters; comprehensive analytical and experimental details can be found in [11].

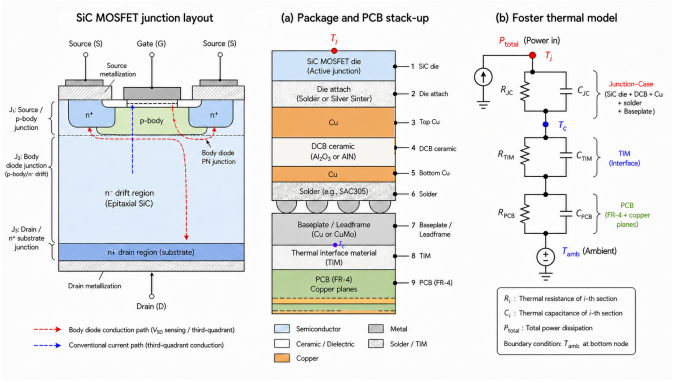


Fig. 2. Cross-sectional physical layer structure of a power semiconductor package alongside its equivalent Foster thermal RC network model.

VII. Electro–Thermal Modeling and Foster Network Comparison

Accurate interpretation of V_{SD}-based junction-temperature measurements in inverter-like operation requires an electro-thermal framework that links the applied power profile to the junction-temperature response as illustrated in Fig. 2. It also provides a basis for quantifying the useful thermal information lost when temperature is reconstructed from slower thermal responses rather than captured near the transient peak [2], [9]. A Foster-type thermal RC network derived from a representative 750 V SiC MOSFET using typical package and transient thermal parameters is therefore used as the reference model for comparison.

This section compares the proposed OVMC + Q_{fw} approach with a slower reading technique to quantify how much useful thermal information is lost and how that loss affects the reconstructed junction temperature and downstream calculations. Although DGB also performs efficiently in transient sensing, its detailed analysis is not pursued here; instead, it is only considered as a relative reference against the OVMC + Q_{fw} based result, since the emphasis of this section is on understanding the information loss caused by the slower technique.

A. Foster-Type Thermal Network

In the Foster representation, the junction-to-case thermal impedance is modeled as a sum of RC branches fitted to the transient thermal impedance $Z_{\mathrm{th}(j-c)}(t)$ [16]:

$$Z_{\mathrm{th}(j-c)}(t) = \sum_{i=1}^{n} R_i \left(1 - e^{-t/\tau_i}\right), \tag{3}$$

where R_i and τ_i denote the thermal resistance and time constant of each branch, respectively. The maximum junction-to-case thermal resistance $R_{\mathrm{th}(j-c,\max)}$ is on the order of 0.23 K/W. Since the cooling phase corresponds to a single current-to-zero transition rather than a repetitive pulse train, the single-pulse $Z_{\mathrm{th}(j-c)}$ curve from the datasheet is used, giving representative values $Z_{\mathrm{th}(j-c)}(100\ \mu\mathrm{s}) \approx 0.02$ K/W and $Z_{\mathrm{th}(j-c)}(10\ \mathrm{ms}) \approx 0.2$ K/W. These values define

the cooling behaviour observed at fast and slow observation windows following the heating pulse.

B. Measured Power Profile and Junction Temperature

Under the dAPC conditions considered here, the device is stressed with a heating current $I_{\text{heat}} = 120$ A at a regulated case temperature $T_c = 80°\text{C}$. Interpolating the on-state resistance to a representative operating point gives $R_{\text{DS(on)}}(100°\text{C}) \approx 9.72\ \text{m}\Omega$, so a pure conduction profile yields $P_{\text{cond}} \approx 140$ W and a realistic worst case dynamic scenario including switching losses gives $P_{\text{total}} \approx 170$ W [5]. Using $R_{\text{th}(j-c,\max)}$, the corresponding peak temperature rises are

$$\Delta T_{\max} = P \cdot R_{\text{th}(j-c,\max)}, \tag{4}$$

which gives $\Delta T_{\max} \approx 32.2°\text{C}$ (DC) and $39.1°\text{C}$ (dAPC), leading to peak junction temperatures of about $112.2°\text{C}$ and $119.1°\text{C}$, respectively.

After the heating pulse, thermal superposition using the single-pulse transient impedances yields cooling drops of

$$\Delta T_{100\mu\text{s}} = P \cdot Z_{\text{th}(j-c)}(100\ \mu\text{s}), \tag{5}$$

which gives $\Delta T_{100\mu\text{s}} \approx 2.8°\text{C}$ (DC), $3.4°\text{C}$ (dAPC), and

$$\Delta T_{10\text{ms}} = P \cdot Z_{\text{th}(j-c)}(10\ \text{ms}), \tag{6}$$

which gives $\Delta T_{10\text{ms}} \approx 28°\text{C}$ (DC), $34°\text{C}$ (dAPC).

A fast window at $100\ \mu$s therefore still captures junction temperatures of approximately $109.4°\text{C}$ and $115.7°\text{C}$, close to the true peak, whereas a conventional 10 ms window observes only about $84.2°\text{C}$ and $85.1°\text{C}$—barely above the regulated case temperature, as shown in Fig. 3. This shows that slow conventional measurement loses nearly all of the useful peak-temperature information, whereas the faster OVMC + Q_{fw} based observation at $100\ \mu$s captures the transient response far more faithfully [1].

C. Package-Induced Losses and Transient Skin Effect

In inverter operation, repeated high-di/dt switching triggers transient skin effects and localized package heating, increasing AC losses and thermomechanical stress [15]. Slow thermal models often miss these short-lived peaks, underestimating transient loss. Package-induced electromagnetic effects in leads and bond wires further alter the electrical waveform and heat injected into the die [14]. In the present setup, the integrated loop stray inductance is assumed to be approximately $L_\sigma \approx 15$ nH and the output capacitance is $C_{\text{oss}} \approx 382$ pF, giving a parasitic ringing frequency of

$$f_{\text{ring}} = \frac{1}{2\pi\sqrt{L_\sigma C_{\text{oss}}}} \approx 66.5\ \text{MHz}. \tag{7}$$

At this frequency, current crowding increases AC resistance and redistributes losses toward the conductor surface. Although the oscillations decay within a few microseconds, the resulting heating is not captured by thermal models evaluated only at $t = 10$ ms. A Foster network fitted to slow thermal responses therefore cannot reproduce the transient peak behavior, while fast V_{SD} sampling at around $100\ \mu$s captures the temperature response more faithfully, although it still represents a window-averaged or lumped temperature rather than the instantaneous peak at the ringing event [1].

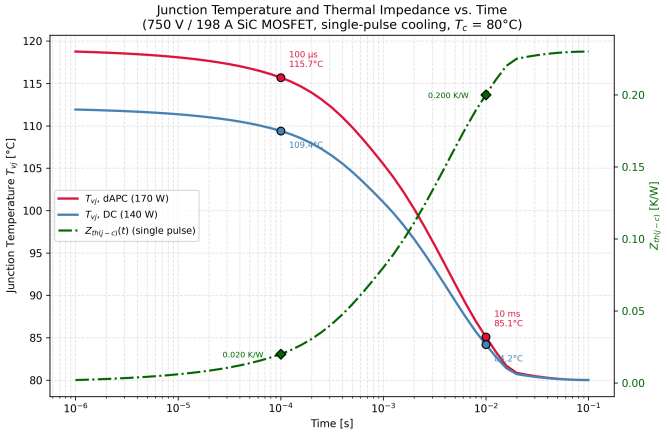


Fig. 3. 3D electro-thermal response illustrating the evolution of virtual junction temperature (T_{vj}) as a function of time and transient thermal impedance ($Z_{th(j-c)}$) for a 750 V, 198 A SiC MOSFET.

VIII. Performance Synthesis and Discussion

The third-quadrant voltage drop (V_{SD}) provides robust virtual junction temperature (T_{vj}) tracking because off-state measurement is less sensitive to gate-drive noise. Reliable operation requires strict sensing conditions, including limiting the current density to approximately $60\ \text{mA/cm}^2$ (about 50–100 mA) and applying a negative gate bias in the range of -6 V to -7 V.

A clear trade-off exists between measurement accuracy and implementation complexity. Hardware-assisted OVMC + Q_{fw} methods achieve sub-Kelvin precision ($< \pm 2-3°\text{C}$) by reducing the commutation delay to about $100\ \mu$s, whereas sensorless Dual-Gate-Bias (DGB) firmware avoids additional hardware but typically incurs a larger error of about $5.0°\text{C}$–$7.5°\text{C}$. Even so, DGB remains attractive because its fast sampling window can better follow rapid thermal effects associated with high-frequency parasitic ringing. Conventional slow measurements based on 10 ms windows can miss approximately $28°\text{C}$–$34°\text{C}$ of cooling, which may cause the power-cycling lifetime to be overestimated. This highlights that slow reconstruction can yield overly optimistic reliability estimates compared with fast VSD-based temperature tracking.

IX. Conclusion and Future Perspectives

Reliable V_{SD}-based T_{vj} estimation depends on fast sampling, strict current-density limits, and sufficiently negative gate bias to avoid large thermal estimation errors. However, the main limitation of bulk V_{SD} sensing is spatial blindness: it cannot resolve localized die hot spots caused by high-frequency transient skin-effect heating. Future work should therefore combine fast, non-invasive electrical V_{SD} measurements with finite-element thermal analysis and machine-learning-based estimators, while also using low-inductance packaging to reduce parasitic ringing.

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